

RailGuard: Physics-Based Simulation and Experimental Design for Railway Wheel-Flat Detection Using Vibration Analysis

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Abstract

A wheel flat is a localized defect on a railway wheel that can create an impact each time the wheel rotates. This paper presents a physics-based simulation of wheel-flat vibration using a two-degree-of-freedom wheel-bogie model with unilateral Hertzian wheel-rail contact. A total of 2,000 simulations were run across ten flat sizes (0–45 mm), four rolling speeds, and independent $\pm 10\%$ variation of five physical parameters. Time- and frequency-domain features were calculated from each simulated acceleration signal. Introducing a flat produced a clear increase in impulse factor and kurtosis relative to the healthy-wheel condition, but neither feature increased monotonically across the complete flat-size range. A preliminary comparison of one representative feature from each method found that STFT spectral entropy had a stronger association with flat size than the selected FFT- and CWT-based features. This feature-level result does not establish the general superiority of STFT over CWT. All reported results are simulation-based. A low-cost benchtop test rig has been designed for future physical validation, but it has not yet been built and no physical measurements are included.

Keywords: wheel-flat detection; vibration analysis; Hertzian contact; Monte Carlo simulation; time-frequency analysis; condition monitoring

1 Introduction

1.1 Problem Statement

A wheel flat is a localized flattened region on the rolling surface of a railway wheel. It can form when a wheel slides during braking or temporarily loses normal rolling contact with the rail. As the wheel rotates, the flat may cause repeated wheel-rail impacts, increased vibration, and elevated dynamic loads that can contribute to wheel, bearing, and rail damage (Wu and Thompson, 2002; Mazilu, 2007; Pieringer and Kropp, 2008).

Early detection is therefore important for railway maintenance. Existing approaches include wayside systems that use fixed trackside instrumentation and onboard approaches that measure vibration closer to the wheel, for example with axle-box accelerometers (Belotti et al., 2006; Mosleh et al., 2021; Mohammadi et al., 2023). If informative vibration features can be identified, a future monitoring system may be able to combine low-cost sensors with edge-computing hardware.

This study investigates that possibility through physics-based simulation. A simplified wheel-bogie dynamic model was developed for signals representing different wheel-flat sizes and speeds. A Monte Carlo dataset of 2,000 simulations was produced by varying five physical parameters. Time- and frequency-domain features were then extracted, including root mean square (RMS), kurtosis, impulse factor, FFT peak frequency, short-time Fourier transform (STFT) spectral entropy, and continuous wavelet transform (CWT) spectral entropy.

The purpose is not to present a completed wheel-flat detector. Instead, the study assesses whether the simulated vibration signals contain measurable changes associated with wheel-flat size, compares selected FFT-, STFT-, and CWT-derived descriptors, and defines a benchtop platform for later experimental validation.

1.2 Related Work

Vibration and acceleration analysis have been widely investigated for railway wheel-defect detection. Liang et al. (2013) studied wheel flats and rail-surface defects using a laboratory roller rig and an axle-box accelerometer. They compared STFT, Wigner–Ville, and wavelet-based representations and showed that wheel-flat impacts produce characteristic time–frequency components that vary with operating conditions.

Wavelet methods are useful for representing short-duration changes in both time and frequency and have been applied to railway wheel-flat diagnosis (Belotti et al., 2006; Jia and Dhanasekar, 2007). However, a time–frequency representation is not automatically the most informative option for every measurement setup or extracted feature. For example, Mohammadi et al. (2023) compared autoregressive, principal-component, and CWT-based feature-extraction methods in a wayside detection framework, while Mosleh et al. (2021) used spectral kurtosis across multiple records. These studies illustrate that performance depends on the defect, sensor location, representation, and downstream analysis.

The physical behaviour of wheel–rail contact and impact has also been studied through models of different complexity (Wu and Thompson, 2002; Sun and Dhanasekar, 2002; Mazilu, 2007; Pieringer and Kropp, 2008). Such models can provide controlled signals before a physical test system is available, although their results must ultimately be checked against measurements.

1.3 Research Gap

Prior studies demonstrate that wheel-flat detection is feasible, but experimental systems often require specialized infrastructure and published datasets may not match the scale or sensing configuration of a low-cost student-built rig. A transparent benchtop-scale simulation can bridge part of this gap by identifying promising signal features, exposing model limitations, and defining measurements that must be collected during later physical validation. The present study therefore combines a reproducible reduced-order simulation with a clearly separated proposal for an unbuilt experimental platform.

1.4 Research Questions

- RQ1. Do simulated axle-box vibration signals show measurable differences as wheel-flat size changes?
- RQ2. How do selected FFT-, STFT-, and CWT-derived features differ in their relationship with wheel-flat size?
- RQ3. How strongly do kurtosis and impulse factor separate simulated healthy and defective wheel conditions?
- RQ4. What measurements and components are needed to validate the simulation with a benchtop test rig?

1.5 Hypotheses and Study Scope

- H1. Wheel-flat size will produce measurable changes in time-domain vibration features, particularly RMS, kurtosis, and impulse factor.
- H2. The selected CWT-based feature will have a stronger relationship with flat size than the selected FFT-based feature because CWT represents changes in both time and frequency.
- H3. A lightweight machine-learning classifier could distinguish wheel-flat conditions using the simulated feature set.

H1 is tested using the Monte Carlo dataset. H2 is evaluated through a preliminary comparison of FFT peak frequency, STFT spectral entropy, and CWT spectral entropy. H3 is not tested because no classifier is trained or evaluated. RQ4 is addressed as an experimental design question; the physical platform has not yet been used to collect measurements. The study therefore concerns simulation and feature analysis, not real-world detector performance.

2 Methods

2.1 Physics-Based Simulation Model

RailGuard uses a simplified two-degree-of-freedom (2-DOF) vertical model. A wheel mass is connected to a bogie mass through a primary suspension represented by a linear spring and damper. The rail is treated as rigid and flat. Related reduced-order and wheel-rail interaction models are described by Sun and Dhanasekar (2002), Liang et al. (2013), and Mazilu (2007). The wheel flat is represented as a geometric change in rolling radius rather than an imposed force multiplier.

Let u_w and u_b denote the vertical displacements of the wheel and bogie relative to static equilibrium. Wheel rotation is described by

$$\theta(t) = \omega t, \quad \omega = \frac{v}{R}, \quad (1)$$

where v is rolling speed and R is wheel radius. The equations of motion are

$$m_w \ddot{u}_w = [F_c(t) - F_{c,\text{eq}}] - k_s(u_w - u_b) - c_s(\dot{u}_w - \dot{u}_b), \quad (2)$$

$$m_b \ddot{u}_b = k_s(u_w - u_b) + c_s(\dot{u}_w - \dot{u}_b). \quad (3)$$

The normal contact force is represented by the unilateral Hertzian law

$$F_c(t) = k_h [\max(\delta(t), 0)]^{3/2}, \quad (4)$$

where k_h is the Hertzian contact coefficient and $\delta(t)$ is the contact deflection. Hertzian normal contact is a standard approximation in contact mechanics and railway interaction modelling (Johnson, 1985; Montenegro and Calçada, 2023). The contact deflection is

$$\delta(t) = \delta_{\text{static}} - u_w(t) - p(\theta(t)). \quad (5)$$

When $\delta > 0$, the wheel and rail are in contact; when $\delta \leq 0$, the contact force is zero. This formulation permits loss of contact and subsequent re-contact without manually prescribed impact states.

For flat length L , the half-angle of the flat is

$$\alpha_h = \arcsin\left(\frac{L}{2R}\right). \quad (6)$$

For angular position ϕ measured from the flat centre, the geometric depth is

$$p(\phi) = \begin{cases} R - \frac{R \cos(\alpha_h)}{\cos(\phi)}, & |\phi| < \alpha_h, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

As the flat enters the contact region, the effective rolling radius decreases. If contact cannot be maintained, the contact force falls to zero before re-contact.

Static equilibrium satisfies

$$k_h \delta_{\text{static}}^{3/2} = (m_w + m_b)g. \quad (8)$$

Table 1. Nominal simulation parameters.

Parameter	Symbol	Unit	Value
Wheel mass	m_w	kg	0.7
Bogie mass	m_b	kg	2.0
Suspension stiffness	k_s	N/m	2×10^6
Suspension damping	c_s	N s/m	5,000
Hertzian contact coefficient	k_h	N/m ^{1.5}	1.5×10^8
Wheel radius	R	m	0.06
Gravitational acceleration (equilibrium only)	g	m/s ²	9.81
Sampling rate	f_s	Hz	10,000
Analysis-window length	T	s	0.6

Each simulation begins at this equilibrium with zero displacement and velocity, avoiding an artificial gravity-related startup transient. The equations are integrated with the adaptive RK45 method using a relative tolerance of 10^{-6} .

The model excludes rail flexibility, distributed contact, lateral wheel motion, suspension nonlinearity, wheelset torsion, creep, and thermal effects. Consequently, simulated amplitudes should not be interpreted as exact predictions of acceleration on a real train. The model is used to determine whether wheel-flat geometry produces consistent feature changes when selected physical parameters vary.

2.2 Signal Processing and Feature Extraction

Each run produced a 0.6-s acceleration signal sampled at 10,000 Hz. The time-domain features were RMS, peak amplitude, crest factor, kurtosis, skewness, shape factor, and impulse factor. RMS describes overall signal magnitude; kurtosis measures concentration in sharp peaks; and impulse factor compares the largest peak with the mean absolute signal level. The definitions used were

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}, \quad (9)$$

$$\text{CF} = \frac{x_{\text{peak}}}{\text{RMS}}, \quad (10)$$

$$\text{IF} = \frac{x_{\text{peak}}}{\text{mean}(|x|)}, \quad (11)$$

$$K = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \mu}{\sigma} \right)^4. \quad (12)$$

FFT was used to obtain the overall frequency content and dominant peak frequency. Welch power spectral density was used to calculate spectral entropy and normalized energy in three bands: 0–200, 200–1,000, and 1,000–3,000 Hz. STFT represented changes in frequency content over time, with time-averaged STFT spectral entropy retained as the representative STFT feature.

CWT was calculated with the PyWavelets implementation (Lee et al., 2019) using the complex Morlet wavelet `cmor1.5-1.0`. Forty-eight logarithmically spaced target frequencies from 50 to 3,000 Hz were used; scales were calculated from the wavelet centre frequency and the 10,000-Hz sampling rate. Energy at each scale was obtained from squared coefficient magnitude and normalized into probabilities p_i . Shannon entropy was then calculated as

$$H = - \sum_i p_i \log_2(p_i). \quad (13)$$

The released dataset contains 14 features: RMS, Peak, CrestFactor, Kurtosis, Skewness, ShapeFactor, ImpulseFactor, SpectralEntropy, three Welch band-energy features, FFT_PeakFreq, STFT_SpectralEntropy, and CWT_SpectralEntropy. The present paper formally analyses a selected subset aligned with the research questions; the remaining variables are retained for reproducibility and future multivariate modelling.

2.3 Monte Carlo Experimental Design

The design contained ten flat sizes (0, 5, 10, $\dots$, 45 mm), four rolling speeds (5, 10, 15, and 20 m/s), and 50 repetitions for each size–speed combination:

$$10 \times 4 \times 50 = 2,000. \quad (14)$$

For every repetition, wheel mass, bogie mass, suspension stiffness, suspension damping, and Hertzian contact coefficient were varied independently by $\pm 10\%$ around their nominal values. The standard deviation of added sensor noise was also sampled between 0.05 and 0.20 m/s². Static equilibrium and contact force were recalculated after parameter randomization. At the lowest speed, the 0.6-s window contained more than eight wheel rotations. Flat-size batches were generated with fixed random seeds in the released scripts.

2.4 Statistical Analysis

Pearson and Spearman correlation coefficients quantified associations between flat size and selected features. One-way analysis of variance (ANOVA) tested whether values differed among the ten flat-size groups. Because a significant omnibus ANOVA only indicates that at least one group differs, adjacent flat-size groups were additionally compared using Cohen’s d effect size (Cohen, 1988). The feature-level comparison used FFT peak frequency, STFT spectral entropy, and CWT spectral entropy. No machine-learning classifier was trained.

2.5 Proposed Experimental Validation Platform

A benchtop platform was designed for future validation but has not been manufactured or used to collect data. A driven aluminium roller with a nominal diameter of 180 mm acts as the rail substitute. An interchangeable 120-mm wheel is pressed against the roller by a wheel-carrier arm. A spring and load cell are planned to control and measure the normal load.

Table 2. Principal components of the proposed, unbuilt test platform.

Component	Reason for selection
ADXL355 accelerometer	Low-noise MEMS sensor with SPI interface for impact-vibration measurement.
Hall-effect sensor	Once-per-revolution speed reference synchronized with vibration data.
Load cell and HX711	Measurement and logging of the normal contact load.
Raspberry Pi 5	Edge processing, data logging, and sensor-channel synchronization.
12-V DC gear motor	Controlled and repeatable roller rotation.

Figure 1 is an intentionally schematic representation. It replaces the earlier AI-generated pseudo-engineering drawing, whose dimensions and bill of materials were not CAD-verified.

Before testing, the roller would be balanced, sensor noise measured, and the Hall sensor and load cell calibrated. The experiment would then compare measured signals with simulated trends under matched flat-size and speed conditions.

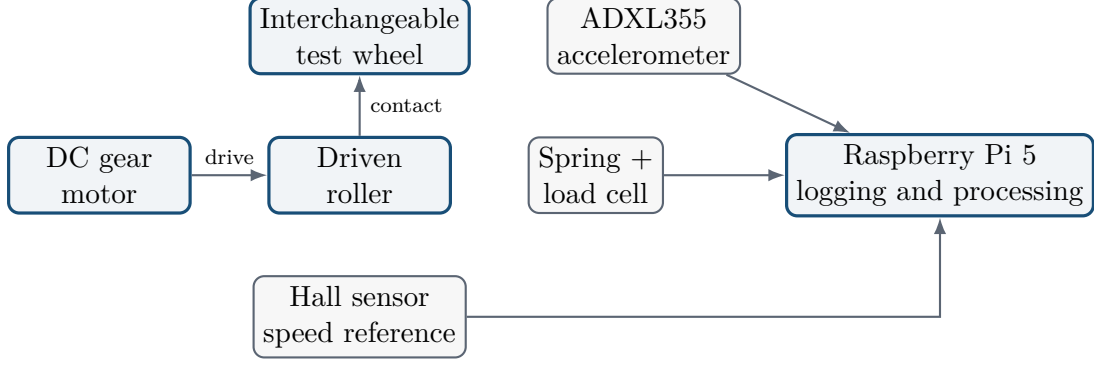


Figure 1. Conceptual layout of the proposed RailGuard benchtop platform (not to scale and not a manufacturing drawing).

3 Results

All results in this section were obtained from the 2,000 simulations described in Section 2. No physical hardware measurements were used.

3.1 Healthy Versus Defective Signal Comparison

Figure 2 compares one healthy-wheel run with one 10-mm-flat run, both at 10 m/s and nominal parameters. The healthy signal remains near zero and is dominated by simulated sensor noise, with a standard deviation of approximately 0.10 m/s^2 and peak magnitude of about 0.39 m/s^2 . The defective signal contains repeated impacts at intervals of approximately 0.038 s, consistent with

$$T_{\text{rot}} = \frac{2\pi R}{v} \approx 0.0377 \text{ s}. \quad (15)$$

Approximately 16 impacts therefore occur in the 0.6-s window, and the largest acceleration in this example is about 34 m/s^2 .

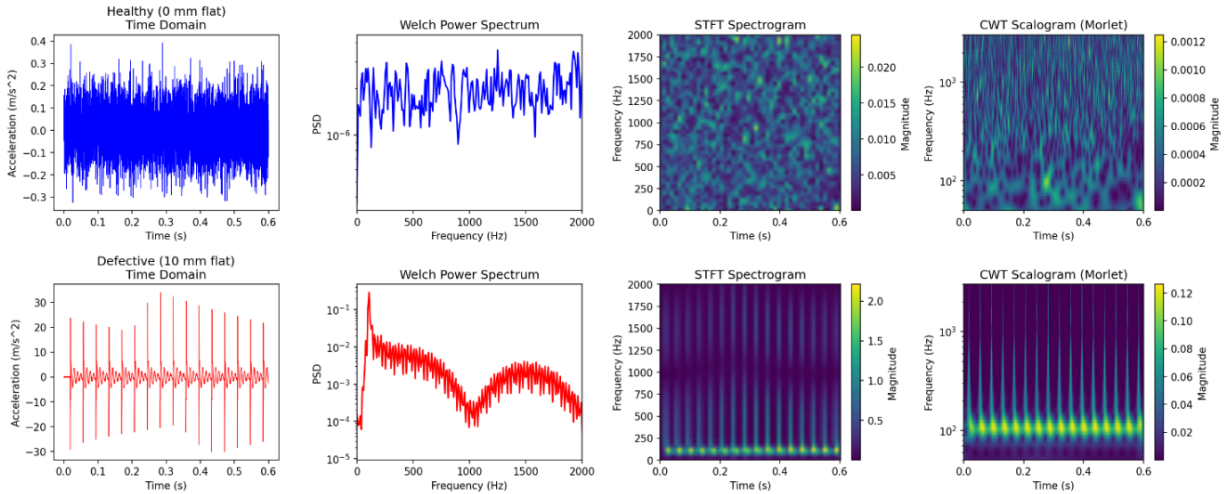


Figure 2. Simulated healthy-wheel (0-mm flat) and defective-wheel (10-mm flat) acceleration at a rolling speed of 10 m/s. From left to right: time-domain signal, Welch power spectrum, STFT spectrogram, and CWT scalogram.

The defective signal contains substantially more spectral energy. The STFT and CWT representations show that this energy recurs at impact times. The CWT scalogram preserves time localization for interpretation, although the CWT descriptor analysed in Section 3.3 summarizes energy over the complete signal.

3.2 Monte Carlo Statistical Validation

Figure 3 and Table 3 summarize RMS, kurtosis, and impulse factor across the ten flat-size groups ($n = 200$ per size).

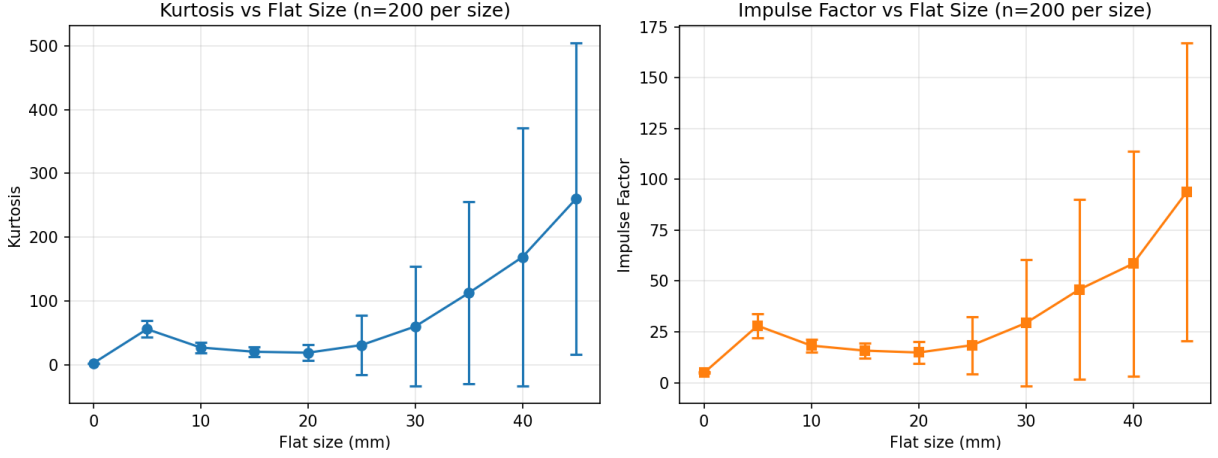


Figure 3. Kurtosis (left) and impulse factor (right) by flat size. Markers show means and error bars show one standard deviation across 200 simulations per size ($N = 2,000$).

3.2.1 Kurtosis

Mean kurtosis was 2.99 ± 0.06 for the healthy condition, close to the theoretical value of 3 for Gaussian noise. It rose to 56.26 at 5 mm, fell to 19.38 at 20 mm, and then increased to 260.55 at 45 mm. The group effect was significant, $F(9, 1990) = 102.28$, $p < .001$ (reported computational value $p = 2.47 \times 10^{-157}$). Pearson correlation was $r = .461$ and Spearman correlation was $\rho = .337$. These moderate coefficients and the group means show that kurtosis did not increase monotonically across the full range. Variability grew sharply at larger sizes, reaching an SD of 243.31 at 45 mm.

3.2.2 Impulse Factor

Mean impulse factor increased from 4.83 ± 0.37 for the healthy condition to 27.99 ± 5.79 at 5 mm, then declined to 14.82 at 20 mm before rising to 93.82 at 45 mm. The group effect was significant, $F(9, 1990) = 122.64$, $p < .001$ (reported computational value $p = 1.54 \times 10^{-183}$). Pearson correlation was $r = .492$ and Spearman correlation was $\rho = .366$. The largest adjacent-group separation was between 0 and 5 mm ($d = 5.64$). Effects were negative for 5–10 mm ($d = -2.09$), 10–15 mm ($d = -0.70$), and 15–20 mm ($d = -0.20$), before becoming positive from 20 to 45 mm ($d = 0.25$ – 0.54). The SD increased to 73.18 at 45 mm.

The Monte Carlo analysis therefore supports measurable separation between healthy and defective simulated wheels but not a simple monotonic mapping from flat size to either kurtosis or impulse factor.

3.3 Comparison of Representative FFT-, STFT-, and CWT-Derived Features

Table 4 compares one representative feature from each signal representation.

STFT spectral entropy had the strongest absolute Pearson correlation ($|r| = .629$), followed by FFT peak frequency ($|r| = .454$) and CWT spectral entropy ($|r| = .305$). The absolute Spearman coefficients were similar for STFT ($|\rho| = .521$) and FFT ($|\rho| = .516$), with a smaller value for CWT ($|\rho| = .241$). Thus, H2 was not supported: the selected CWT descriptor did not outperform the selected FFT descriptor. This limited feature-level comparison neither establishes that STFT is generally superior to CWT nor measures classification accuracy.

Table 3. Feature mean and standard deviation by flat size ($n = 200$ per size).

Flat size (mm)	RMS mean $\pm$ SD	Kurtosis mean $\pm$ SD	Impulse factor mean $\pm$ SD
0	0.123 ± 0.043	2.99 ± 0.06	4.83 ± 0.37
5	2.384 ± 0.269	56.26 ± 12.96	27.99 ± 5.79
10	3.716 ± 0.549	27.46 ± 8.51	18.22 ± 3.16
15	5.397 ± 1.721	20.84 ± 7.44	15.75 ± 3.85
20	7.614 ± 3.656	19.38 ± 12.40	14.82 ± 5.53
25	10.349 ± 7.133	31.35 ± 46.14	18.43 ± 13.99
30	18.016 ± 20.707	60.45 ± 93.55	29.31 ± 31.07
35	28.734 ± 29.854	112.96 ± 142.73	45.86 ± 44.19
40	32.806 ± 30.520	169.16 ± 202.25	58.52 ± 55.23
45	59.262 ± 52.539	260.55 ± 243.31	93.82 ± 73.18

Table 4. Association of representative signal features with flat size ($N = 2,000$).

Feature	Pearson r	Spearman ρ	ANOVA F
FFT peak frequency	-.454	-.516	518.1
STFT spectral entropy	-.629	-.521	506.7
CWT spectral entropy	-.305	-.241	205.9

Note. All omnibus ANOVA tests had $p < .001$.

4 Discussion

4.1 Interpretation of Results

The primary result is strong separation between the healthy condition and the first defective group. At 5 mm, mean impulse factor increased from 4.83 to 27.99 and mean kurtosis from 2.99 to 56.26; the adjacent-group effect sizes were $d = 5.64$ and $d = 5.81$, respectively. Within this model, a small flat therefore produces a readily measurable signal change.

However, neither feature changes monotonically. Both decline from 5 mm to approximately 20 mm before increasing at larger sizes. Their means alone are consequently insufficient for reliable estimation of flat size across 0–45 mm. Moreover, increasing standard deviations indicate strong sensitivity to the Monte Carlo parameter combinations. The nonlinear contact formulation may contribute to this variability, but the simulation cannot determine whether the pattern reflects real wheel–rail behaviour or only properties of the reduced-order model.

The healthy-wheel kurtosis of 2.99 provides a basic consistency check because the healthy signal is dominated by Gaussian background noise rather than a large artificial startup transient. The signal-representation comparison further shows why a representation should be evaluated through the features actually extracted from it. Here, STFT spectral entropy had the strongest Pearson association and CWT spectral entropy the weakest, but alternative wavelets, frequency ranges, or time-localized descriptors could change that ordering.

4.2 Comparison With Related Work

The periodic impacts and altered frequency content are consistent with experimental and modelling studies of wheel-flat vibration (Liang et al., 2013; Wu and Thompson, 2002; Mazilu, 2007). Previous research also shows that different diagnostic pipelines may favour different representations: wavelet approaches can be effective (Belotti et al., 2006; Jia and Dhanasekar, 2007), while autoregressive or spectral-kurtosis methods may be more informative in other settings (Mohammadi et al., 2023; Mosleh et al., 2021). The present results reinforce the narrower conclusion that CWT should not

be assumed superior without a common feature and evaluation framework.

Unlike experimental studies, the present work uses controlled simulation and Monte Carlo parameter variation. This helps isolate model behaviour and generate a reproducible dataset, but it cannot substitute for physical validation. The proposed test platform is intended to determine which trends persist when actual mechanical tolerances, contact behaviour, and sensor noise are introduced.

4.3 Limitations

1. No physical measurements were collected, and no real-world detection performance is claimed.
2. Kurtosis and impulse factor were non-monotonic; the evidence supports healthy-versus-defective separation more strongly than direct flat-size estimation.
3. Feature variability increased substantially at larger flat sizes, and the current model cannot show whether the same behaviour will occur experimentally.
4. The 2-DOF model omits rail flexibility, distributed contact, car-body and wheelset dynamics, creep and traction forces, suspension nonlinearity, and thermal effects.
5. The FFT-STFT-CWT comparison used one feature per representation and did not apply a common classifier or regression model.
6. The CWT descriptor reduces scale-wise energy to one entropy value and does not retain the complete time-localized coefficient field for classification.
7. The nominal wheel mass is 0.7 kg; the manufactured wheel may differ and must be measured before validation.
8. Although 14 features are included in the released dataset, only the subset directly relevant to the stated research questions is analysed here.

4.4 Future Work

The next stage is to construct and calibrate the benchtop platform. The actual wheel mass and contact load should be measured, the roller balanced, and the accelerometer, Hall sensor, and load-cell chain calibrated. Repeated measurements should then be collected under matched flat-size and speed conditions and compared with the simulated signals and feature trends.

Subsequent analysis should evaluate several descriptors from each signal representation within a common classification and regression framework. A transparent threshold baseline should precede lightweight models such as random forests. Recent work using a 1:10-scale test rig and sequence models provides one possible experimental comparator (Cui et al., 2025). If sufficient physical data become available, more complex models may be considered, but they should be evaluated with held-out experimental runs rather than simulation alone. Differences between measured and simulated behaviour should be used to refine the contact and structural assumptions.

5 Conclusion

RailGuard used a 2-DOF wheel-bogie model with unilateral Hertzian contact to generate 2,000 simulated acceleration signals across ten flat sizes, four speeds, and Monte Carlo variation of five physical parameters. Healthy and defective conditions were clearly separated: mean impulse factor increased from 4.83 at 0 mm to 27.99 at 5 mm, and mean kurtosis increased from 2.99 to 56.26. Nevertheless, neither feature increased monotonically across the entire flat-size range, and both became more variable at larger sizes.

Among the three representative frequency-domain features, STFT spectral entropy had the strongest Pearson association with flat size ($r = -.629$), compared with FFT peak frequency ($r = -.454$) and CWT spectral entropy ($r = -.305$). The selected CWT feature therefore did not support H2, but the comparison does not establish general superiority of one representation. Overall, the simulation supports vibration-based separation of healthy and defective conditions

within the model; it does not establish reliable flat-size estimation or field performance.

Table 5. Status of the study hypotheses.

Hypothesis	Result	Status
H1	Healthy and defective conditions were clearly separated, but selected features were not monotonic across flat size.	Partly supported
H2	STFT spectral entropy had the strongest association; the selected CWT feature had the weakest.	Not supported
H3	No classifier was trained or evaluated.	Not tested

6 Reproducibility

The project materials include the final 2,000-row Monte Carlo dataset, simulation and feature-extraction code, statistical-analysis scripts, and figure-generation workflow. The dataset contains 14 extracted features and simulation metadata. CWT processing uses PyWavelets with `cmor1.5-1.0`, 48 logarithmically spaced frequencies from 50 to 3,000 Hz, and the entropy calculation described in Equation 13. Fixed random seeds are assigned in the released scripts. To permit exact independent reproduction, the final submission package should additionally state the repository or archive URL and the Python, NumPy, SciPy, and PyWavelets versions used.

AI-Assisted Figure Disclosure

The earlier manuscript contained an AI-generated concept image with unverified engineering dimensions. That image has been removed from this version. Figure 1 is a schematic diagram created directly in LaTeX and contains no manufacturing dimensions or CAD claims. Figures 2 and 3 display outputs from the simulation and statistical-analysis workflow.

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